

ROHAN KUMAR

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Professional Summary

Production-focused **Agentic AI Engineer** with **2.5+ years of experience** owning the end-to-end full-stack development of AI systems (Agentic, Search, Rule-based, etc.) to enhance product offerings and optimize company workflows in a **fast-paced EdTech startup**. Expert in building **intelligent automation** and **multi-agent orchestration frameworks**.

Technical Skills

Languages: Python, SQL, C, JavaScript/TypeScript

Cloud & Infra: AWS (EC2, S3), GCP (Cloud Run, Storage, VertexAI), Docker, Linux

Frameworks & Tools: FastAPI, Next.js, React, PostgreSQL, Redis, LangSmith, CI/CD, Git

AI: ML, NLP, GenAI (Prompt Engineering, Agentic Engineering, Multi-Agent Orchestration, Fine-Tuning: SFT/DPO/GRPO, RAG), First-Order Logic, Vector Databases

Experience

AI Researcher

July 2023 – Present

Cerebry (IPREP Learning Solution)

Noida, India

- **AI Architect & Lead** — Spearheaded end-to-end full-stack development of AI platforms, owning architecture, model fine-tuning, and production rollout.
- **GradingEngine:** Fine-tuned **Kimi-k2.5** on custom datasets using **GRPO** to achieve precise scoring based on rubrics, and **DPO** to deliver enhanced multi-lingual, personalized feedback, achieving **80% accuracy** on custom benchmarks.
- Researched and developed **QuestionCrafter:** A **full-stack, collaborative deep multi-agent system** that clarifies requirements, conducts deep research, and executes **long-horizon asynchronous tasks (running up to 3 hours independently)** to generate and edit 1000+ high-complexity, curriculum-aligned questions with **90% accuracy**. Boosted efficiency by **10x**, eliminated manual coding, and cut content operational costs by **50%** with same-day delivery.
- Built **TweakMath:** First-Order Logic-driven multi-agent generator of pedagogically equivalent math variants—**~90% cheaper than our prior industry-leading Rule-Based AI System**, minute-level delivery speeds, reduced SME dependency; powers our adaptive, anti-cheating practice platform.

Software Engineer Intern

Nov 2022 – May 2023

FantomStarter

Remote

- Developed **SoloKadenaMiner**, a lightweight software to do **ASIC mining on Kadena blockchain**.
- Optimized ASIC communication & achieved about 50% reduction in share rejection rate.
- Used .NET Core, Socket Programming, Asynchronous & Reactive Programming, and Stratum protocol.

Academic & Personal Projects

LLM Stream Processor | [Repo](#) | [PyPI](#)

May 2025 – Dec 2025

- Developed a high-performance library for **callback-driven real-time sanitization** of LLM streams, ensuring **prefix-safe** PII redaction and content moderation with **5μs overhead** per token.
- Implemented an Aho-Corasick automaton for multi-pattern matching, enabling **async and background actions** on sensitive data patterns within real-time synchronous and asynchronous streams.

Vision | [Repo](#) | [Demo](#)

Feb 2023 – Apr 2023

- A personal **assistant for visually impaired people** on Android.
- **Users can double-tap and speak in natural language.** The system **automatically captures images** to provide **scene explanations, live object detection**, answer visual questions like medicine expiry dates, and manage tasks.
- Features an agentic architecture, where an **LLM (text-only) controls vision models** for OCR, Image Captioning, Zero-Shot Object Detection, Live Object Detection, & dynamically generates SQL queries for task management.
- Used GPT-3.5 Turbo, TF-Lite (on-device inference), FastAPI, PyTorch, Android, Docker, Transformers Library, Text encoders, & Vector database.

Education

University of Calcutta

Kolkata, India

Bachelor of Technology in Information Technology; CGPA: 8.41/10

2019 – 2023

M.S. Memorial Public School

New Delhi, India

Class XII (AISSCE); Aggregate: 82.2%

2016 – 2018